Public repository copy – filtered, not the complete submission draft. Tissue images, individual-slide plots, case-level tables and row-level source data are withheld. Marked omissions preserve the original numbering and captions. References to Source Data describe the local research archive; only screened aggregate tables are distributed here. Reported aggregate estimates are unchanged.

Supplementary information

Foundation-model-based whole-slide classification of salivary gland MALT lymphoma and Sjögren's syndrome

Supplementary Methods

S1 Hybrid-Evidence MIL

Task formulation and feature projection. Each WSI was represented by a sampled feature bag $X = \{\mathbf{x}_i\}_{i=1}^{N}$ and its associated patch coordinates (a_i, b_i), where $N = 512$ and $\mathbf{x}_i \in \mathbb{R}^{1536}$. The slide label was $y = 1$ for MALT and $y = 0$ for SS. No patch-level labels were used. A shared projection placed all features in a $D = 256$ dimensional hidden space:

$$\mathbf{h}_i = \mathrm{Dr}_{0.3}[\mathrm{GELU}(\mathrm{LN}(\mathbf{W}_p\mathbf{x}_i + \mathbf{b}_p))], \qquad \mathbf{h}_i \in \mathbb{R}^D. \tag{S1}$$

Here LN denotes layer normalization and $\mathrm{Dr}_{0.3}$ denotes dropout with probability 0.3 during training; dropout is disabled during inference. All vectors below are column vectors, and $[\cdot\,;\cdot]$ denotes concatenation along the feature dimension.

Gated attention evidence. Gated pooling followed the attention mechanism of Ilse and colleagues [1], with an additional value transformation implemented in the archived model. For a sequence $Z = \{\mathbf{z}_j\}_{j=1}^{L}$ of D dimensional vectors, define the pooling operator $\mathcal{G}_\theta$ as

$$\mathbf{t}_j = \mathrm{Dr}_{0.3}[\mathrm{ReLU}(\mathbf{W}_v\mathbf{z}_j + \mathbf{b}_v)], \tag{S2a}$$

$$\ell_j = \mathbf{w}_a^{\mathsf{T}}\left[\tanh(\mathbf{V}\mathbf{t}_j + \mathbf{b}_V) \odot \mathrm{sigmoid}(\mathbf{U}\mathbf{t}_j + \mathbf{b}_U)\right] + b_a, \tag{S2b}$$

$$\alpha_j = \frac{\exp(\ell_j)}{\sum_{l=1}^{L}\exp(\ell_l)}, \qquad \mathcal{G}_\theta(Z) = \sum_{j=1}^{L}\alpha_j\mathbf{t}_j. \tag{S2c}$$

The symbol $\odot$ denotes element-wise multiplication. All attention and value matrices map between D dimensional spaces. The attention evidence vector was $\mathbf{e}_{\mathrm{att}} = \mathcal{G}_{\theta_A}(\{\mathbf{h}_i\}_{i=1}^{N})$. Thus, the pooling weights multiply transformed values $\mathbf{t}_i$. The regional branch used the same operator with independently learned parameters θ_R.

Distribution evidence. The center and heterogeneity of the sampled bag were summarized by two separate vectors:

$$\mathbf{e}_\mu = \frac{1}{N}\sum_{i=1}^{N}\mathbf{h}_i, \tag{S3a}$$

$$\mathbf{e}_\sigma = \left[\frac{1}{N-1}\sum_{i=1}^{N}(\mathbf{h}_i - \mathbf{e}_\mu)^{\odot 2}\right]^{\odot 1/2}. \tag{S3b}$$

Powers and square roots in Eq. (S3) are element-wise. The sample denominator $N - 1$ matches the standard deviation used in the code.

Prototype-margin evidence. Two latent prototype banks indexed by $c \in \{0, 1\}$ each contained $K = 8$ learnable prototypes $\mathbf{p}_{c,r} \in \mathbb{R}^D$. Let a hat denote the implemented ℓ_2 normalization, with its numerical safeguard for zero norms. The maximum cosine similarity within each bank and the bank-1-minus-bank-0 margin were

$$s_{i,c} = \max_{1 \le r \le K} \widehat{\mathbf{h}}_i^{\mathsf{T}} \widehat{\mathbf{p}}_{c,r}, \qquad m_i = s_{i,1} - s_{i,0}, \tag{S4a}$$

$$k = \min(32, N), \qquad I_+ = \mathrm{TopKIdx}(\{m_i\}, k), \qquad I_- = \mathrm{TopKIdx}(\{-m_i\}, k), \tag{S4b}$$

$$\mathbf{e}_+ = \frac{1}{k} \sum_{i \in I_+} \mathbf{h}_i, \qquad \mathbf{e}_- = \frac{1}{k} \sum_{i \in I_-} \mathbf{h}_i. \tag{S4c}$$

TopKIdx returns indices of the k largest entries. The high- and low-margin vectors therefore average the instances with the largest and smallest margins, respectively; the selected values need not all have the same sign. Prototypes were optimized through differentiable similarity and margin statistics in the slide-level classification objective. The hard top-k index selections were discrete; no additional prototype-specific loss or regional labels were used. The bank indices do not identify validated MALT or SS morphology, and a larger margin is not constrained to increase the final MALT score.

Coordinate-ordered groups and low-margin evidence. To form coordinate-ordered groups, let π be the permutation obtained by sorting the implemented coordinate key $10^6 b_i + a_i$. The ordered hidden sequence was partitioned using the code's chunk operation into $Q \le 16$ consecutive non-empty chunks C_j (16 equal chunks for the 512 sampled patches). Group tokens and the combined low-margin/regional vector were

$$\mathbf{r}_j = \frac{1}{|C_j|} \sum_{l \in C_j} \mathbf{h}_{\pi(l)}, \qquad j = 1, \ldots, Q, \tag{S5a}$$

$$\mathbf{e}_{\mathrm{reg}} = \mathcal{G}_{\theta_R}(\{\mathbf{r}_j\}_{j=1}^{Q}), \qquad \mathbf{e}_{\mathrm{nr}} = \mathbf{e}_- + \mathbf{e}_{\mathrm{reg}}. \tag{S5b}$$

The regional operator includes its own value transformation from Eq. (S2a). Coordinates determine group membership, but the subsequent pooling is invariant to the order of the group tokens. The branch has no explicit intergroup distances, adjacency or positional encoding. Equal-sized groups of sampled instances were not anatomical regions and can span different physical extents.

Similarity and margin statistics. Eight scalar statistics supplemented the evidence vectors. Define $T_c = \mathrm{TopKIdx}(\{s_{i,c}\}, k)$, $\bar{s}_c^{\mathrm{top}} = k^{-1} \sum_{i \in T_c} s_{i,c}$ and $\bar{m} = N^{-1} \sum_i m_i$. Then

$$s_m = \left[\frac{1}{N-1} \sum_{i=1}^{N} (m_i - \bar{m})^2 \right]^{1/2}, \tag{S6a}$$

$$\rho_0 = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(m_i > 0), \qquad \rho_{\bar{m}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(m_i > \bar{m}), \tag{S6b}$$

$$\mathbf{u} = \left[\bar{s}_0^{\mathrm{top}}, \bar{s}_1^{\mathrm{top}}, \bar{m}, s_m, \max_i m_i, \min_i m_i, \rho_0, \rho_{\bar{m}} \right]^{\mathsf{T}} \in \mathbb{R}^8. \tag{S6c}$$

$\mathbb{I}$ is the indicator function. The top-similarity sets T_c are computed separately for the two banks and need not coincide with the margin-based sets I_+ and I_-.

Learned gating, weighted concatenation and classification. The five evidence vectors were ordered as $(\mathbf{e}_1, \ldots, \mathbf{e}_5) = (\mathbf{e}_{\mathrm{att}}, \mathbf{e}_\mu, \mathbf{e}_\sigma, \mathbf{e}_+, \mathbf{e}_{\mathrm{nr}})$. A two-layer gate received the unweighted vectors and statistics:

$$\mathbf{q} = [\mathbf{e}_1; \mathbf{e}_2; \mathbf{e}_3; \mathbf{e}_4; \mathbf{e}_5; \mathbf{u}] \in \mathbb{R}^{5D+8}, \tag{S7a}$$

$$\boldsymbol{\gamma} = \mathrm{softmax}(\mathbf{W}_{g2}\, \mathrm{Dr}_{0.3}[\mathrm{GELU}(\mathbf{W}_{g1}\mathbf{q} + \mathbf{b}_{g1})] + \mathbf{b}_{g2}) \in \mathbb{R}^5, \tag{S7b}$$

$$\widetilde{\mathbf{q}} = [\gamma_1 \mathbf{e}_1; \gamma_2 \mathbf{e}_2; \gamma_3 \mathbf{e}_3; \gamma_4 \mathbf{e}_4; \gamma_5 \mathbf{e}_5; \mathbf{u}], \tag{S7c}$$

$$\mathbf{z} = \mathrm{Dr}_{0.3}[\mathrm{GELU}(\mathrm{LN}(\mathbf{W}_f \widetilde{\mathbf{q}} + \mathbf{b}_f))], \tag{S7d}$$

$$a = \mathbf{w}_o^{\mathsf{T}}\, \mathrm{Dr}_{0.3}[\mathrm{ReLU}(\mathrm{LN}(\mathbf{W}_c \mathbf{z} + \mathbf{b}_c))] + b_o, \qquad p = \mathrm{sigmoid}(a). \tag{S7e}$$

The gate dimensions were $\mathbf{W}_{g1} \in \mathbb{R}^{D\times(5D+8)}$ and $\mathbf{W}_{g2} \in \mathbb{R}^{5\times D}$; $\mathbf{W}_f \in \mathbb{R}^{D\times(5D+8)}$ and $\mathbf{W}_c \in \mathbb{R}^{D\times D}$. Each γ_j is non-negative and $\sum_j \gamma_j = 1$. Equation (S7c) preserves five weighted feature blocks; the eight statistics enter fusion without multiplication by a branch weight. The gate supplies sample-dependent feature scaling, rather than a distribution of causal contributions. Branch norms, the learned fusion transformation and the ungated statistics also affect the classifier. The output p is an uncalibrated MALT model score in $[0, 1]$.

Training objective. For a minibatch $\mathcal{B}$ of slides, the weighted binary cross-entropy was

$$\mathcal{L} = -\frac{1}{|\mathcal{B}|}\sum_{b\in\mathcal{B}} \left[w_+ y_b \log p_b + (1-y_b)\log(1-p_b)\right], \qquad w_+ = \frac{77}{67}. \tag{S8}$$

The implementation evaluated this objective directly from logits for numerical stability. The positive-class weight was the ratio of the 77 SS to 67 MALT training slides. Class weighting changes the optimum of binary cross-entropy; the sigmoid output was not assumed to equal an unweighted posterior probability. No calibration model was fitted to either evaluation set. Optimization and the archived checkpoint are described below.

S2 Historical model settings and exploratory ensembles

The retained Hybrid checkpoint contains 1,521,672 parameters and was selected at epoch 15 from a 30-epoch run with batch size eight. The archived ABMIL contains 6,690,306 parameters, retains a 1,536-dimensional representation and was selected at epoch 37 from a 50-epoch run with batch size 16. Both use learning rate 5×10^{-5} and weight decay 10^{-4}. The classifier and attention dimensions differ; these comparisons evaluate complete implementations. The checkpoint inventory in Source Data supplies exact dimensions, training budgets and identities for the inspected models.

The local CLAM-style implementation omits the original instance-clustering loss. The local two-tier attention model uses partitioned bags and two aggregation stages; the transformer variant uses a conventional transformer and coordinate multilayer perceptron. The grouped-region transformer averages coordinate-ordered groups before transformer processing. The latter is not the published R^2T feature re-embedding architecture [2]. Prototype-pooling names describe local implementations and are not assigned an unsupported correspondence to a specific published ProtoMIL method. Table S1 retains all historical results under these descriptive names; the code-ID mapping accompanies the data.

The historical ensemble was explored by enumerating combinations of candidate predictions and ranking them by internal AUROC and AP. Hybrid, local CLAM-style pooling and the grouped-region transformer were combined by equal probability averaging. Its internal AUROC was 0.9649. At threshold 0.5 it had 21 true positives, 22 true negatives, four false positives and two false negatives. The separately explored threshold 0.70 yielded 21, 25, one and two, respectively, and accuracy 93.88%. Both member selection and threshold exploration used development outputs; this is not a prespecified independent validation result. The historical three-model ensemble was not evaluated externally.

Earlier compound configurations and seed experiments are preserved in the historical Source Data. Enabling the high-margin vector also enabled eight statistics, and low-margin/regional components were added together. Moreover, training patch samples used an unseeded NumPy generator in that component study. Its results cannot independently attribute performance to each module. Earlier seeds 7, 13 and 21 used 20 epochs, whereas seed 42 used 30, with differing partitions. They are not pooled with the later controlled seeds.

S3 Controlled training and algorithm screen

The first matched experiment used attention-only, full fixed scaling and full learned gating, each with seeds 17, 29 and 43. Attention-only retained the shared projection and attention/classifier dimensions; fixed scaling multiplied each of the five vectors by 0.2 while retaining the statistics and fusion layer. Their parameter counts were 658,178, 1,190,403 and 1,521,672. Within each seed, shared tensors were identically initialized. Exact training bags and minibatch orders were matched, but masks after divergent computation branches were not assumed to be elementwise identical.

The later screen trained ten configurations with the same three seeds (30 runs), then the chosen candidate and Hybrid with seeds 59 and 71 (four runs). The attention and Hybrid controls reproduced the first matched

experiment's selected epochs and classifications, with maximum score discrepancy 5.96×10^{-8}. Thus, the studies include nine prior runs and 34 later runs, with six deliberately repeated controls, not 43 independent initialization settings.

Common settings were 30 epochs, AdamW with learning rate 5×10^{-5}, weight decay 10^{-4}, batch size eight, positive loss weight 77/67 and dropout 0.3. The first strictly highest validation-AUROC epoch was retained. There was no scheduler, early stopping, probability calibration or threshold optimization. The following deviations define the screen:

- **Input L2 normalization:** each 1,536-dimensional patch vector was divided by its norm (safeguard 10^{-12}) and multiplied by $\sqrt{1536}$ before Hybrid or matched attention. This changes input magnitude as well as removing norm variation.
- **Attention masking:** six randomly chosen logits among the top ten attention logits were masked during training. Evaluation used unmasked attention. This is an STKIM-inspired component adaptation, not full ACMIL [3].
- **Original-space attention:** gated attention with hidden dimension 256 pooled the original 1,536-dimensional values, followed by a linear classifier; dropout was 0.25.
- **Feature-subspace attention:** a shared sorted random subset of 256 feature coordinates was used for attention within each training minibatch, while values retained all 1,536 dimensions. At inference, logits from 21 contiguous 256-coordinate windows with stride 64 were averaged before sigmoid. A separate configuration used batch size 32 and learning rate 3×10^{-4}, changing both optimization factors and the update count (150 versus 540). These are nnMIL-inspired adaptations under the local 30-epoch protocol [4].
- **Two-bag training:** classification losses from two independently sampled 512-patch bags were averaged. The consistency variant added 0.1 times symmetric Bernoulli KL divergence. For logits z_1, z_2, this was implemented as $\frac{1}{2}\,\mathrm{mean}[(\sigma(z_1) - \sigma(z_2))(z_1 - z_2)]$, with gradients through both paths. Patch samples and dropout both varied; this extends the perturbation beyond original R-Drop [5]. The two-bag classifier without KL was the corresponding control.

The eight modified candidates were ranked by mean validation AUROC over seeds 17, 29 and 43, rounded to 12 decimal places. Ties used smaller sample SD, then parameter count and configuration ID. The selected Attention+L2 configuration was frozen before further seed training and new external inference. The predefined engineering target was an internal mean AUROC increase of at least 0.01 over Hybrid, with at least two non-decreasing paired seeds; a mean balanced-accuracy drop exceeding 0.02 was flagged. The observed increase was 0.0011, so the target was not met. Added seeds and external results did not trigger reselection. All 34 runs are in Table S5; the complete protocol is included with the source bundle.

Raw-mean and normalized-mean logistic controls were fitted to full-feature slide means from the training set. Training-only standardization and fixed logistic hyperparameters prevented evaluation slides from contributing to fitted feature scaling. Their 512-patch and full-feature inference results are separate. The former uses the common reference bags, whereas the latter uses every available feature row (Table S6).

S4 Sampling rules, reproducibility and statistics

Training bag seeds were derived from the first four SHA-256 bytes, interpreted as an unsigned little-endian 32-bit integer. Shared controlled-run seeds encoded the training seed, epoch and slide code; order and dropout used separate keys. Two-view training used a distinct second-view key. Exact indices and orders were archived. The original reference evaluation retained full external-manifest indices even after the final subset was selected. For historical fixed-checkpoint resampling, 20 new per-slide bags used keys encoding study date, cohort, publication-coded slide and repeat. Sampling was without replacement unless fewer than 512 rows existed; historical internal indices were sorted when sampling without replacement. The later five-bag study used a distinct fixed key, so those five bags exclude the reference bag.

Probability ensembles averaged every specified seed and bag; no seed was omitted. The subspace model's within-model feature-window *logit* averaging is distinct from between-model probability averaging. Bootstrap

comparisons used aligned resamples across models. The paired external difference for five-seed/five-bag Attention+L2 minus Hybrid was −0.0164 (95% interval −0.0448 to 0.0058). Its difference from the historical primary checkpoint was −0.0407 (−0.1320 to 0.0267); these are different comparators, not interchangeable baselines. All intervals are conditional on the saved predictions.

Primary Hybrid, ABMIL and other reference predictions were reproduced before the comparative analyses. An independent CPU audit of the later screen verified all 8,246 main prediction rows and 76 metric rows, and checked six predictions per neural checkpoint (204 in total; maximum CPU/GPU discrepancy 1.79×10^{-7}). Sampling indices, train-only linear standardization and ensemble calculations were independently checked. These are numerical-reproducibility checks, not additional clinical validation.

The analyses used Python 3.10.18, NumPy 1.26.4, pandas 2.3.2, SciPy 1.15.3, scikit-learn 1.7.2 and Matplotlib 3.10.6. Later inference/training used PyTorch 2.8.0 and CUDA 12.8 on an NVIDIA A800 80GB. The 34-run algorithm workflow, including loading and evaluation, took 620.9 seconds and peaked at 4.70GiB of PyTorch-allocated GPU memory; it reused cached features. This is a local shared-hardware observation, not an independently benchmarked runtime. The historical training runtime remains completion item T02.

S5 Acquisition and image-region audit

Header-derived scan software was NDP.scan 2.5.85 for 133 internal slides and 3.3.0 for 60; all 168 analyzed external slides recorded 2.5.85. Extraction attributes recorded zero overlap and a 1,024-pixel level-0 patch footprint. These acquisition fields do not determine clinical provenance or actual stain. The inspected TRIDENT checkout identifies version 0.2.0 and revision prefix a6305acfef68. Full local file identities are retained in the project provenance record; the exact encoder-weight revision remains T01.

The four illustrated model cases comprise a MALT true positive, SS true negative, SS false positive and MALT false negative. Across 12 visualization cases, the largest absolute performance–visualization score difference was 0.3252, without a threshold classification change. Margin colors were normalized independently per slide using a symmetric scale based on the 2nd and 98th percentiles, so color intensity is not directly comparable across slides. The mean-feature gate was the largest on all 217 reference bags; a gate is a feature-scaling coefficient, not a contribution decomposition.

All eight displayed thumbnails matched their original 1,024-pixel windows after the archived resize to 220 pixels. The corresponding field width was approximately 231.1 μm, supporting 50-μm bars. Automated segmentation coverage was 100% for seven windows and 48.11% for the low-margin window in the false-negative case. Polygon holes were included by the archived segmentation convention, so these are geometric mask overlaps, not verified tissue fractions. For that case, archived overlay geometry used a slide height of 62,208 pixels, whereas the current reader reported 70,400; the original geometry is preserved. Its near-empty-looking crop has not been replaced or interpreted as a mechanism of failure. Pathology review and geometry resolution remain completion item H06. The separate illustrative photomicrographs lack original physical calibration and are labeled accordingly.

Supplementary Tables

Table S1: Historical complete-implementation comparison on 49 validation slides

Implementation	AUROC	AP	Sens. (%)	Spec. (%)
Three-model ensemble	0.9649	0.9576	91.30	84.62
Hybrid	0.9599	0.9514	95.65	80.77
ABMIL	0.9498	0.9396	82.61	88.46
Local two-tier attention	0.9398	0.9416	86.96	88.46
Local attention pooling (CLAM-style)	0.9365	0.9422	91.30	84.62
Local grouped-region transformer	0.9281	0.8400	95.65	84.62
Local prototype pooling	0.9264	0.9306	86.96	84.62
Mean pooling	0.9247	0.9332	78.26	92.31
Local transformer pooling	0.9231	0.9180	65.22	96.15
Local dual-stream pooling	0.9197	0.9260	78.26	92.31
Distribution–prototype pooling	0.9130	0.9178	91.30	69.23
CONCH v1.5 + TITAN	0.9013	0.9203	91.30	73.08

Threshold 0.5. This table retains all historical models, not a ranking of official full-method reproductions. TITAN uses a different encoder and instance budget. Code-ID mappings and original metrics are in Source Data. The ensemble combines Hybrid, local CLAM-style pooling and the grouped-region transformer.

Table S2: External complete-checkpoint comparison on identical reference bags

Implementation	AUROC	Sens. (%)	Spec. (%)	TP/FN/ TN/FP
Hybrid	0.8721	93.42	68.75	142/10/11/5
ABMIL	0.7903	42.76	81.25	65/87/13/3
Local attention pooling (CLAM-style)	0.6591	58.55	62.50	89/63/10/6
Local grouped-region transformer	0.7743	89.47	50.00	136/16/8/8

Table S3: Three-seed controlled training comparison

Configuration	Parameters	Validation AUROC	External AUROC
Matched attention	658178	0.9348 ± 0.0044	0.7270 ± 0.1035
Full fixed scaling	1190403	0.9398 ± 0.0153	0.7445 ± 0.1254
Full learned gate	1521672	0.9353 ± 0.0070	0.7633 ± 0.1581

Seeds 17, 29 and 43. Means ± sample SD. The algorithm screen reran the attention and learned-gate controls and reproduced them; those reruns are not new independent seeds.

Table S4: All ten configurations in the three-seed algorithm screen

Configuration	Validation AUROC	External AUROC
Hybrid control	0.9353 ± 0.0070	0.7633 ± 0.1581
Matched attention	0.9348 ± 0.0044	0.7270 ± 0.1035
Hybrid + L2	0.9337 ± 0.0140	0.7166 ± 0.2016
Attention + L2	0.9365 ± 0.0060	0.7726 ± 0.1111
Attention masking	0.9348 ± 0.0073	0.7356 ± 0.0940
Original-space attention	0.9041 ± 0.0463	0.6051 ± 0.2023
Feature-subspace attention	0.8701 ± 0.0174	0.7876 ± 0.0617

Configuration	Validation AUROC	External AUROC
Subspace, batch32/lr3e-4	0.8790 ± 0.0070	0.7707 ± 0.0504
Hybrid, two bags	0.9337 ± 0.0245	0.6886 ± 0.2175
Hybrid, two bags + KL	0.9337 ± 0.0235	0.7307 ± 0.1694

Table S5: All 34 neural training runs: selected checkpoint results

Configuration	Seed	Epoch	Val. AUC	Ext. AUC	Ext. sens.	Ext. spec.
Hybrid control	17	29	0.9298	0.7311	60.53	68.75
Matched attention	17	13	0.9314	0.8170	80.26	62.50
Hybrid + L2	17	26	0.9264	0.6571	74.34	56.25
Attention + L2	17	19	0.9314	0.8976	73.03	93.75
Attention masking	17	13	0.9314	0.8100	75.00	62.50
Original-space attention	17	30	0.8980	0.5391	52.63	50.00
Feature-subspace attention	17	29	0.8562	0.7788	50.00	75.00
Subspace, batch32/lr3e-4	17	28	0.8813	0.7722	54.61	75.00
Hybrid, two bags	17	8	0.9114	0.6525	62.50	56.25
Hybrid, two bags + KL	17	8	0.9114	0.6727	59.21	56.25
Hybrid control	29	20	0.9331	0.9350	59.21	93.75
Matched attention	29	18	0.9331	0.7500	61.18	62.50
Hybrid + L2	29	18	0.9247	0.9412	65.79	93.75
Attention + L2	29	18	0.9348	0.7352	60.53	62.50
Attention masking	29	18	0.9298	0.7669	51.97	68.75
Original-space attention	29	26	0.9532	0.4441	28.95	50.00
Feature-subspace attention	29	11	0.8896	0.7307	78.95	50.00
Subspace, batch32/lr3e-4	29	4	0.8846	0.7196	78.95	37.50
Hybrid, two bags	29	15	0.9298	0.9219	83.55	87.50
Hybrid, two bags + KL	29	10	0.9314	0.9215	71.71	93.75
Hybrid control	43	29	0.9431	0.6238	17.76	68.75
Matched attention	43	12	0.9398	0.6139	34.87	56.25
Hybrid + L2	43	26	0.9498	0.5514	22.37	56.25
Attention + L2	43	21	0.9431	0.6850	45.39	62.50
Attention masking	43	22	0.9431	0.6299	61.18	56.25
Original-space attention	43	2	0.8612	0.8322	95.39	43.75
Feature-subspace attention	43	7	0.8645	0.8532	88.16	56.25
Subspace, batch32/lr3e-4	43	30	0.8712	0.8203	63.82	68.75
Hybrid, two bags	43	12	0.9599	0.4914	21.71	50.00
Hybrid, two bags + KL	43	28	0.9582	0.5979	20.39	68.75
Hybrid control	59	20	0.9515	0.8236	44.08	87.50
Attention + L2	59	22	0.9214	0.8409	48.68	93.75
Hybrid control	71	27	0.9164	0.9145	64.47	93.75
Attention + L2	71	24	0.9398	0.8717	50.66	93.75

Sensitivities and specificities are percentages at threshold 0.5. Every planned run is retained. Full epoch histories and checkpoints remain in the experiment archive.

Table S6: Deterministic linear controls

Features	Input	Cohort	AUROC	Sens. (%)	Spec. (%)
Raw mean	512 patches	internal	0.9314	78.26	92.31
Raw mean	Full features	internal	0.9264	78.26	92.31

Features	Input	Cohort	AUROC	Sens. (%)	Spec. (%)
L2 mean	512 patches	internal	0.9247	73.91	92.31
L2 mean	Full features	internal	0.9247	73.91	92.31
Raw mean	512 patches	external	0.8335	45.39	93.75
Raw mean	Full features	external	0.8396	45.39	93.75
L2 mean	512 patches	external	0.8462	46.05	93.75
L2 mean	Full features	external	0.8475	44.74	93.75

Table S7: Probability ensembles of the complete seed sets

Model	Averaging	Cohort	AUROC	Sens. (%)	Spec. (%)
Attention + L2	3 seeds; ref. bag	external	0.7817	55.26	68.75
Attention + L2	3 seeds; ref. bag	internal	0.9431	82.61	88.46
Hybrid control	3 seeds; ref. bag	external	0.7730	40.13	81.25
Hybrid control	3 seeds; ref. bag	internal	0.9482	91.30	88.46
Attention + L2	5 seeds; ref. bag	external	0.8199	53.29	81.25
Attention + L2	5 seeds; ref. bag	internal	0.9365	82.61	92.31
Hybrid control	5 seeds; ref. bag	external	0.8470	47.37	93.75
Hybrid control	5 seeds; ref. bag	internal	0.9515	91.30	92.31
Attention + L2	5 seeds; 5 bags	external	0.8314	53.95	87.50
Attention + L2	5 seeds; 5 bags	internal	0.9381	91.30	88.46
Hybrid control	5 seeds; 5 bags	external	0.8479	48.03	93.75
Hybrid control	5 seeds; 5 bags	internal	0.9548	91.30	92.31

Reference bags and the five newly sampled bags are separate inputs. Each displayed metric is calculated after probability averaging; it is not an average of AUROCs.

Table S8: Twenty-bag sensitivity analysis of the retained Hybrid checkpoint

Cohort	Median score SD	Crossing 0.5	Draw AUROC range	Mean-score AUROC
internal	0.0120	7/49	0.9498–0.9682	0.9632
external	0.0150	18/168	0.8549–0.8943	0.8775

Table S9: Linked performance and visualization outputs

Individual-case rows are omitted from this public repository copy.

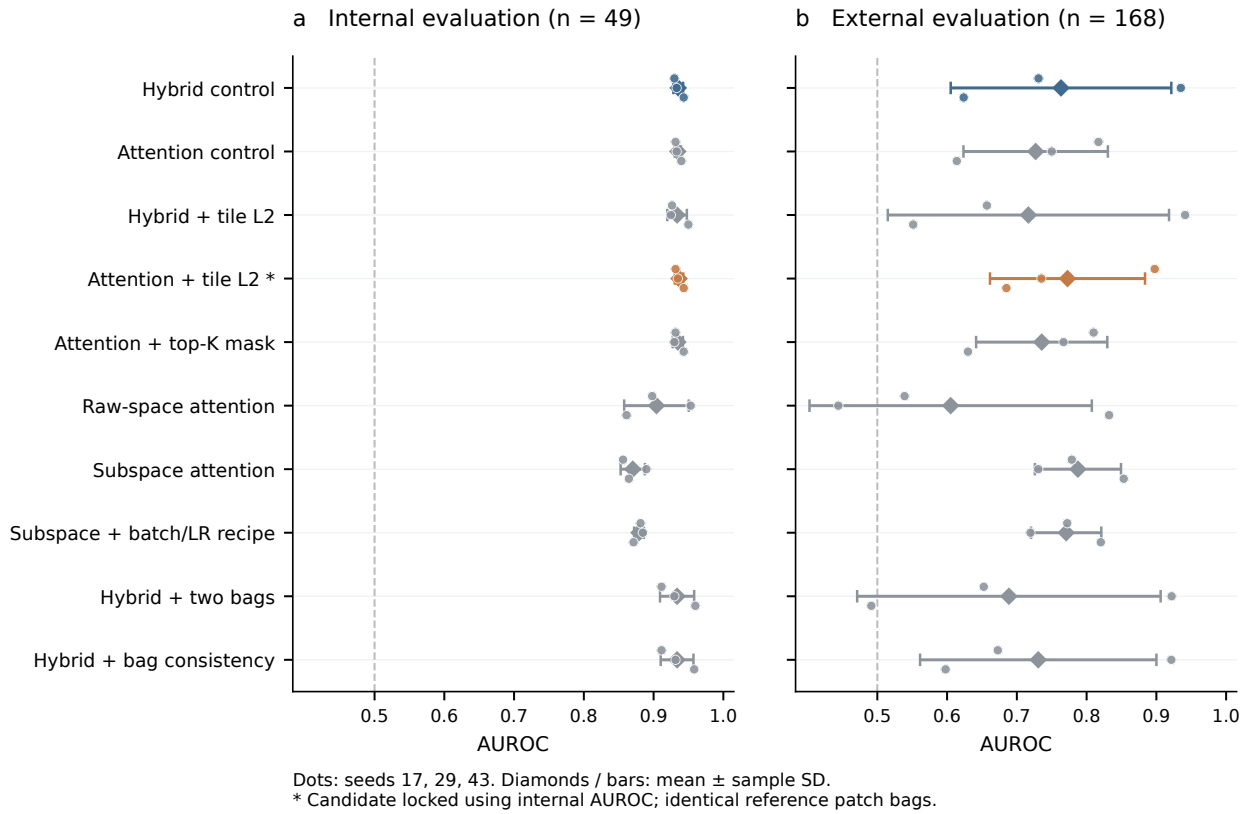


Figure S1: Complete algorithm screen. All ten configurations and all initial seeds (17, 29, 43) are shown for validation and external evaluation. Individual points are separately trained models; bars summarize mean and sample SD, not confidence intervals. Attention+L2 was selected on mean validation AUROC before external inference. Each cohort contains the same 49 or 168 slides across configurations.

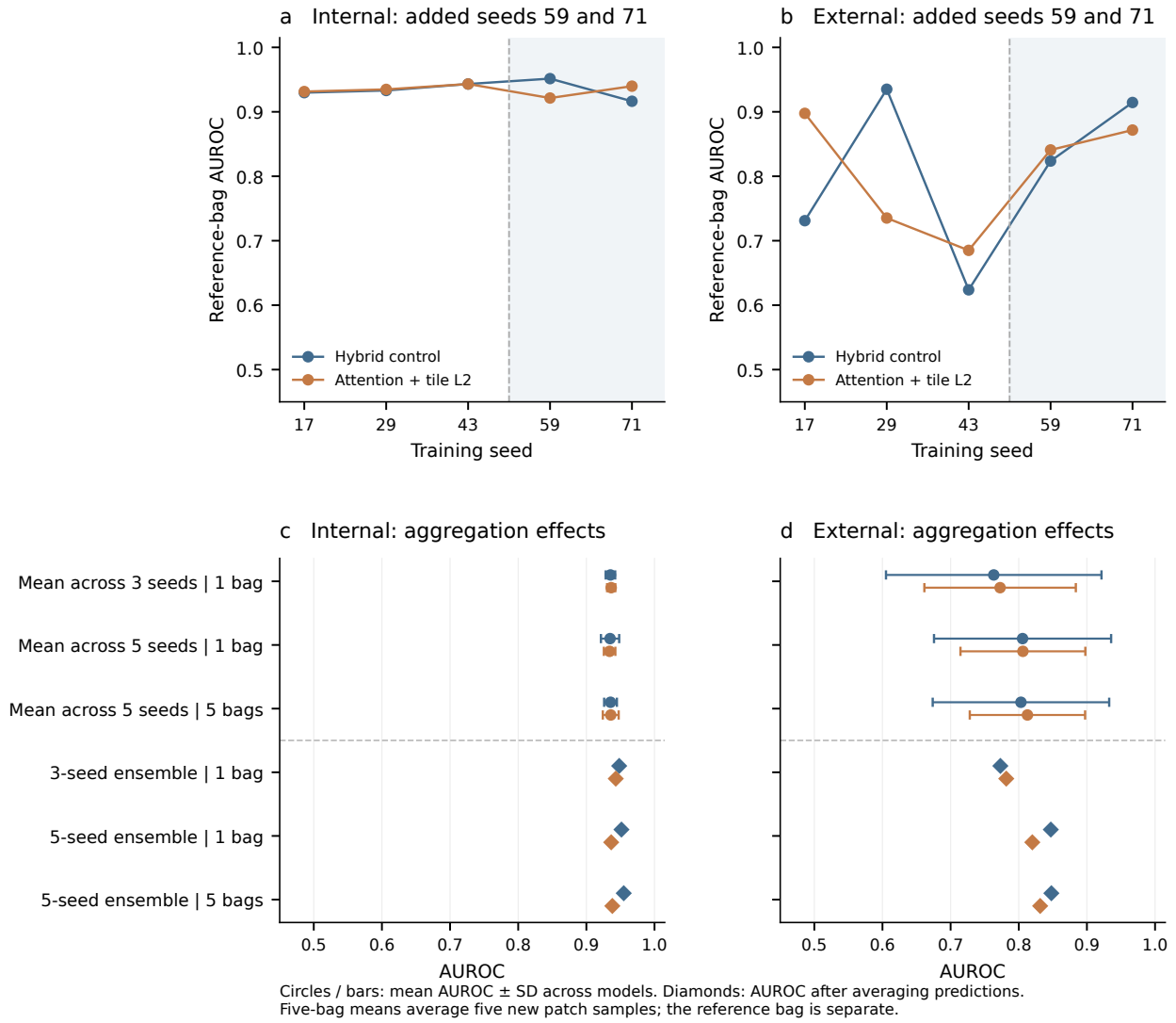


Figure S2: Training seeds and probability aggregation. The top panels show all five seeds for the selected candidate and Hybrid; shaded regions identify seeds 59 and 71. Lower panels distinguish mean AUROC across fitted models (circles and SD bars) from AUROC after averaging their probabilities (diamonds). Five-bag means use five new inputs and exclude the reference bag. No seed was omitted.

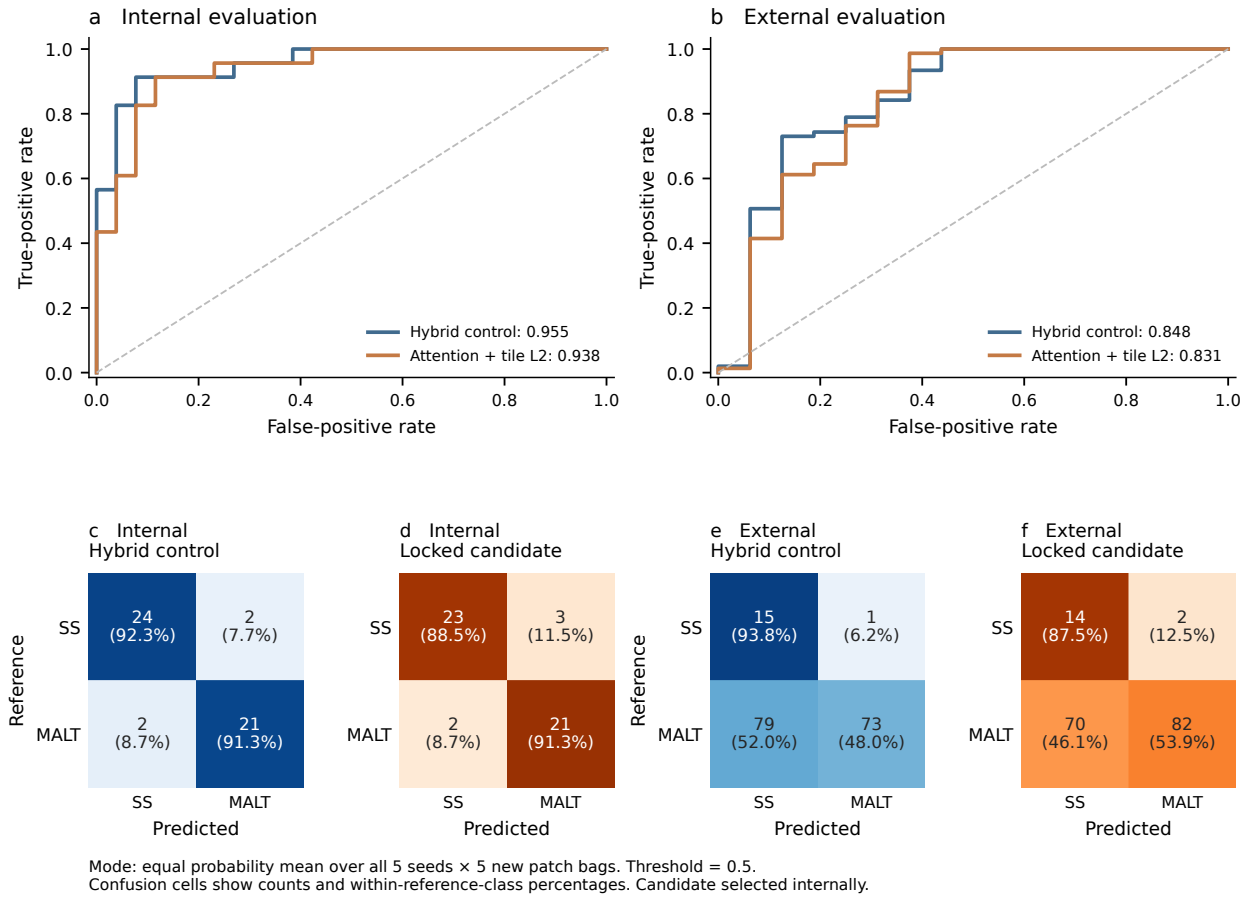


Figure S3: Five-seed, five-bag probability ensembles. All 25 probabilities per slide were averaged for each model before calculating ROC and threshold metrics. Confusion matrices use threshold 0.5 and display counts with within-reference-class percentages. These are supplementary development ensembles, not replacements for the retained primary checkpoint.

Omitted from this public repository copy

This figure contains tissue images or individual-slide observations.
The complete figure is retained in the local manuscript archive.

Figure S4: Archived model-region displays for pathological review. Four cases represent the four threshold outcomes. Performance and visualization bags differed; paired scores for all 12 cases are in Table S9. Colors represent within-slide-normalized latent-bank margins, not verified MALT/SS labels. The eight displayed crops retain their source pixels, with 50-μm bars. The false-negative case has unresolved legacy overlay geometry and a low-mask-coverage window, detailed in Supplementary Methods. These displays are retained as review material; they do not establish disease localization or explain errors.

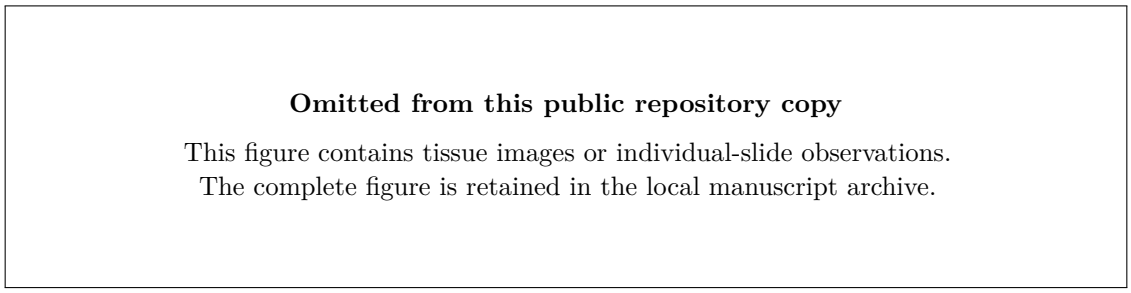


Figure S5: Supplied histological context images. These illustrative MALT and SS images are distinct from the model-selected regions and were not used as regional ground truth. Original pixels are unchanged. Specimen linkage, physical calibration and final descriptions remain completion item H06; no new physical scale is inferred.

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